

Samgar Nurlan

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EDUCATION

University of Waterloo

Candidate of Bachelor of Software Engineering

Waterloo, ON

2025 – 2030

EXPERIENCE

Software Developer

Jun 2026 – Sep 2026

The J Healthcare Initiative

Waterloo, ON

- Collaborated in a six-person development team with a nonprofit client to design and deliver a centralized data-management solution based on real organizational requirements.
- Translated stakeholder feedback into technical requirements and contributed to the design, development, testing, and deployment of database and backend components using C#, .NET, Azure SQL, and Microsoft Azure.
- Strengthened client-facing software development skills by participating in requirements discussions, iterative development, technical documentation, and delivery of a functional prototype for future organizational use.

Software Developer

2025 – Present

UWaterloo Robotics Team

Waterloo, ON

- Improved robot teleoperation usability by implementing WASD keyboard controls, enabling faster operator input and smoother navigation by modifying ROS2 teleop_twist_keyboard nodes.
- Increased runtime control flexibility by adding dynamic velocity tuning, allowing operators to adjust linear and angular speed in real time through ROS2 command interfaces.
- Ensured reliable robot command delivery by validating velocity publishing to /cmd_vel topics, improving control stability through ROS2 debugging and testing tools on Linux environments.

Technical Workshop Lead

2023 – 2025

ACM @ CCNY

New York City, NY

- Educated 20+ students in programming fundamentals by designing and delivering workshops on Python, Data Structures & Algorithms, and Web Development.
- Strengthened industry awareness for students by coordinating guest talks with alumni engineers, helping participants understand real-world software development practices.

PROJECTS

J Healthcare Initiative - Centralized Data Management Prototype

Client-Based Full-Stack Project

C#, .NET, Azure SQL, REST APIs

- Collaborated in a team of six to design a centralized data-management prototype for a nonprofit organization focused on preventing medical overdoses.
- Helped translate stakeholder needs into technical requirements and designed database and intake-form components for collecting funding, program, and operational information.
- Developed and updated the Azure SQL database structure and supported its integration with the .NET backend and data-intake workflow.

Form-ulation - AI Bicep Curl Form Analyzer

Computer Vision & Machine Learning

Python, YOLOv8, OpenCV, Streamlit, React, Vite, Snowflake

- Built a real-time computer vision system that analyzes bicep curl form using YOLOv8 pose estimation and engineered biomechanical features
- Built a scalable ML pipeline integrating pose extraction → feature engineering → classification, enabling real-time inference.
- Implemented webcam-based inference with OpenCV and a Streamlit web app providing feedback and confidence scores

PayGuard – AI-Powered Business Fraud Prevention Platform

Full-Stack Web Application

AI, React, JavaScript, REST APIs

- Built a fraud-prevention platform for businesses to identify and flag potentially suspicious invoices and transactions before payment.
- Developed an AI-powered invoice checker that analyzes invoice details for suspicious patterns, including unusual amounts, bank-account changes, urgent payment requests, and vendor inconsistencies.
- Implemented automated risk scoring and approval workflows that classify invoices by fraud risk and help businesses review high-risk payments before processing.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, C#, SQL, Java, Verilog

Frameworks & Web: React, Vite, Flask, .NET, REST APIs, Streamlit, YOLOv8

Databases: MySQL, SQLite, Snowflake, Azure SQL

Libraries & Tools: Azure, OpenCV, bcrypt, PyJWT, Git/GitHub, GitLab, Docker, ROS2